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Key Technologies for Green Communications: Overview, Opportunities and Challenges

Huaicheng Li

***Abstract*—The arrival of the sixth generation of mobile communications technology (6G) heralds a revolutionary change in connectivity, delivering unprecedented data rates, ultra-low latency, and massive device integration to support transformative applications such as holographic communications and the Internet of Everything. However, this vision also brings with it a key challenge: increasing energy consumption. Achieving a truly sustainable 6G future will require a paradigm shift towards green communications technologies, aiming to maximise network performance while reducing environmental impact. In this paper, we provide an overview of key green communication technologies. Several mainstream green communication key technologies are introduced and some representative research results are listed. Finally, open research challenges and future directions for green 6G communications are discussed.**

***Index Terms*—6G, green communications, communication technologies, lightweight.**

I. Introduction

THE advent of 6G communication networks promises unprecedented data rates, ultra-low latency, and massive connectivity, paving the way for transformative applications like holographic communications, extended reality, and the Internet of Everything [1]. However, realizing this vision comes at a cost: a substantial increase in energy consumption. As the world grapples with the urgent need for sustainable solutions, green communication has emerged as a critical research area, focusing on minimizing the environmental footprint of these next-generation networks [2]. Achieving the goals of 6G while simultaneously mitigating its environmental impact requires a paradigm shift in how we design, deploy, and manage communication infrastructure. This necessitates the development and implementation of innovative green communication technologies across multiple layers, from network optimization and intelligent resource allocation to advanced coding schemes and lightweight data processing techniques [3].

Green communication refers to a communication network that is sustainable through the use of various technologies and strategies to reduce the energy consumption and environmental impact of the communication network [2], [4]. It involves optimising the network architecture, improving resource utilisation, reducing data transmission delays and protecting the environment. Through these technical means, an efficient and environmentally friendly communication network can be built, paving the way for the future 6G era.

While significant progress has been made in developing various green communication technologies, including network optimization [5], intelligent resource allocation [6], and data processing techniques, substantial challenges remain. Current research efforts often face trade-offs between performance gains and energy efficiency. For example, optimizing network operations for minimal energy consumption may impact latency or throughput. Similarly, implementing distributed computing paradigms can reduce data transmission overhead but introduce complexities in coordination and security [7]. Moreover, the increasing heterogeneity and dynamism of 6G networks demand more sophisticated and adaptive algorithms for resource management and energy optimization. Furthermore, the practical implementation and scalability of these technologies in real-world deployments remain key areas of ongoing research.

The overall framework is shown in Fig. 1. This review explores key enabling technologies that hold the potential to unlock a truly sustainable 6G future, including but not limited to network optimization, intelligent scheduling, distributed learning, green coding, and lightweight techniques. Our contributions are as follows:

- An overview of typical cases of key technologies of green communications according to the application models.
- Analyses and outlooks according to the problems still faced by each of the current green communications applications in 6G.

II. Key Technologies for Green Communications

A. Intelligent scheduling

1) Dynamic resource allocation: Dynamic resource allocation is a technology that adjusts the resource allocation strategy in real time according to the network load. Through intelligent scheduling algorithms, dynamic adjustment of resource allocation can ensure that network resources are reasonably utilised, improve system efficiency and reduce unnecessary energy consumption. The general Optimization expression in a dynamic resource allocation task is

$$\begin{aligned} &\max_{\mathcal{S},\mathcal{F}} \sum_{j\in\mathcal{A}} \sum_{i\in\mathcal{N}} s_i^j u_i^j, \\ &s.t. C1: f_i \geq 0, \forall i \in \mathcal{N}, \\ &C2: s_i^j = \{0,1\}, i \in \mathcal{N}, \\ &\ldots, \\ &C_n: \ldots, \end{aligned} \tag{1}$$

where s_i^jusually is the selection strategy, u_i^j usually is the utility function, $C1$ and $C2$ are some constraints on the

Huaicheng Li is with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: huaichengli@bjtu.edu.cn).

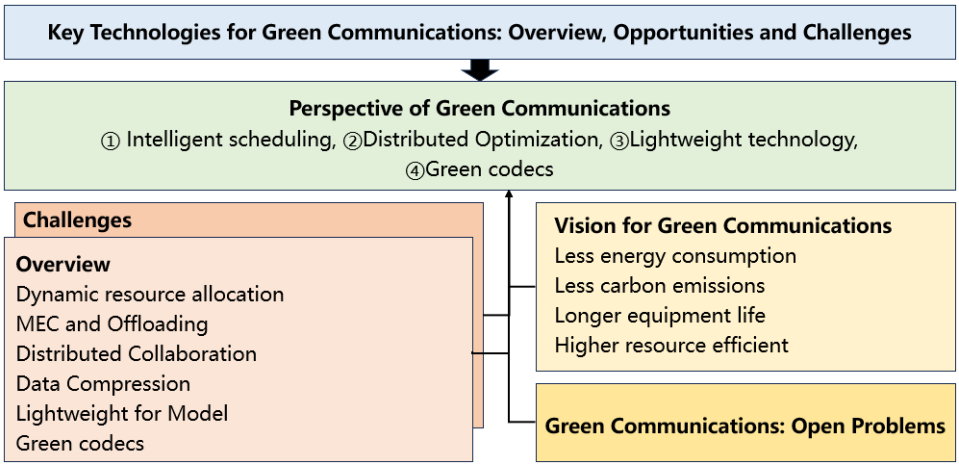

Fig. 1. Overall framework of this review.

system. The Optimization objective is to maximise the utility of the system under the maximum delay constraint of the task.

Traditional resource allocation methods often rely on fixed rules and algorithms, which are difficult to adapt to low-latency and high-reliability environments. artificial intelligence (AI) methods are often used to solve such Optimization problems. AI can achieve more flexible and efficient resource allocation by learning and adapting to user behaviours and the state of the network. For example, AI can dynamically adjust the allocation of spectrum resources based on real-time network load and user demand, thereby improving spectrum utilisation efficiency. Yang et al [8] proposed an Optimization problem based on inter-user joint clustering, sub-channel assignment and successive cancellation for solving the network interference management problem in view of the complex network topology and serious interference caused by the increase in the number of base stations and their small size. Zhao et al [9] designed an adaptive joint resource allocation scheme based on Deep Reinforcement Learning for uplink and downlink computational resource allocation in response to the problem that computationally intensive vehicular tasks are difficult to execute efficiently in resource-constrained vehicular networking scenarios. Xiao et al [10] propose a deep reinforcement learning-based task sequencing and resource allocation algorithm to proficiently manage computational and communication resources in IoV environments to ensure efficient execution of various tasks.

2) Mobile Edge Computing and Offloading: Mobile Edge Computing (MEC) is a technology that places computing resources close to the user, and computation offloading refers to offloading computing tasks from the mobile device to the edge server for processing. In this way, data transfer latency and device energy consumption can be significantly reduced. Zhao et al [11] proposed a task offloading scheme based on the task preprocessing depth deterministic policy gradient algorithm, considering the characteristics of high-speed train operation, and designed the corresponding environment interaction model by using the intelligent body and interaction training, and obtained the optimised task offloading and resource allocation scheme. Nie et al [12] proposed a centralised multi-intelligent body reinforcement learning algorithm for the joint Optimization problem of resource allocation, user association and power control in UAV MEC systems.

B. Distributed Optimization

Network services are becoming smarter with the widespread use of AI models in communication networks. However, the training of these models requires large amounts of data. Federated learning (FL), a distributed machine learning technique, allows multiple devices to train models in their local environment and aggregate the updated model parameters. This approach not only enables model training while protecting user data privacy, but also reduces energy and bandwidth consumption during data transmission.

The general Optimization objective for federated learning is

$$\min_{w} F(w) := \sum_{n \in U} \mu_n \alpha_n F_n(w) \tag{2}$$

where U is the number of users participating in the federated learning architecture, α_n is the weight of each user parameter when aggregated by the central server, and $\mu_n \in \{0, 1\}$ is the selection indicator of the users in the region. F_n is the loss value of each user. Based on this Optimization problem, researchers have proposed a series of studies for high accuracy and low latency requirements. Ma et al [13] investigated how to use RIS to assist in improving the training performance of federated learning systems on a network of vehicular columns, where multiple columns train a shared federated learning model. Multi-row collaboration can alleviate the data processing pressure of a single row with limited computational resources. RIS can enhance inter-row communication in a cost-effective and energy-efficient manner. Quan et al [14] designed a centralised computational offloading and distributed resource allocation framework to reduce the communication overhead between multiple intelligences and used FL techniques to protect user privacy and accelerate training.

There are also some examples of research using model compression techniques to optimise the FL training process. Chang et al [15] proposed a low-complexity joint adaptive

pruning and vehicle scheduling algorithm to optimise and accelerate the sub-deployment training of AI models in FL architectures. Chen et al [16] designed an edge device selection strategy and used model compression to study energy-efficient FL in edge intelligent networks with limited computational and communication. Song et al [17], in order to solve the Optimization problem in a distributed manner, combined federated learning and deep reinforcement learning to propose a distributed sidechain resource allocation method to maximise the network spectrum energy efficiency.

C. Lightweight technology

1) Data Compression: Data compression is a key technique to reduce energy consumption and bandwidth usage by reducing the amount of data transmitted. Reducing the amount of data transmitted through compression algorithms can significantly reduce bandwidth requirements and energy consumption during transmission. The general expression for the Optimization objective of data compression is

$$\max \frac{|X|}{|C|}, \tag{3}$$

$$\min T_{com} + T_{decom}. \tag{4}$$

Need to maximise compression ratio vs. minimise compression and decompression time. Where X denotes the size of the original data and C denotes the size of the compressed data. The traditional sense of data compression is for data lossless compression and decompression, representative algorithms are Hoffman coding [18], [19], dictionary coding [20] and so on.

Another emerging technology is semantic communication. Semantic communication is a communication technology that centres on the idea of transmitting not just sequences of bits of information, but the actual meaning or semantics represented by those sequences of bits. It is often implemented based on Encoder-Decoder architecture. In recent years, semantic communication has attracted wide interest and achieved certain results, among which Joint Source Channel Coding (JSCC) is considered as a key technology [21], [22]. Lin et al [23] proposed a new channel-transferable semantic communications framework, which compresses the information to be transmitted by semantic feature extraction of the information at the sender's end, and then decodes and recovers it at the receiver's end, and the codecs learnt on one type of channel are adapted to other types of channels.

2) Lightweight for Model: Lightweight for Model consists of two main techniques, one is to directly design smaller scale lightweight AI models, and the other is to use model compression algorithms to secondary process AI models to make them lightweight. As 6G ubiquitous intelligence advances, more and more AI models are being used in 6G networks. However, this also brings additional computation and data transfer to the network. Model compression is a technique to reduce energy consumption and storage requirements by reducing the size of AI models. Reducing the number of AI model references by compressing the model can reduce the energy consumption and storage demand of computational resources, thus improving system efficiency and energy saving.

For lightweight model design, Wei et al [24] proposed a novel lightweight general-purpose intrusion detection system for monitoring botnet at the edge of IoT. Convert packet length sequences into 3-channel RGB images based on lightweight CNNs for malicious traffic classification. An et al [25] studied the channel state feedback problem for massive multiple-input multiple-output (MIMO) in FDD mode and proposed LWNet, a lightweight network that integrates a multiresolution encoder and full-dimensional dynamic convolution in the decoder , for enhanced feature extraction. Yan et al [26] designed an efficient ultra-lightweight deep learning network with fairly low computational complexity as a classifier for automatic composite modulation classification in order to adapt to resource-constrained practical applications.

For model compression, Wang et al [27] proposed a lightweight deep learning-based automatic modulation classification method, which is lightweighted using model pruning techniques to make AI models with smaller model sizes and faster computation speeds. Specifically a scaling factor is introduced for each neuron in the CNN and scaling factor sparsity is enforced by compression sensing. It helps to filter out redundant neurons and then prune those neurons. Gao et al [28] developed an ultra-large-scale MIMO channel network to obtain a priori knowledge of channel estimation. The channel estimation network is lightened by weight pruning and quantization to accelerate model inference and facilitate model storage and transmission. Guo et al [29] proposed a FL training method based on matrix low-rank approximation for wireless systems to reduce the communication cost. Al-Ahmadi et al [30] proposed an innovative deep learning based user device localisation method for massive MIMO systems that utilises reconfigurable RIS to enhance the localisation performance. They employed a knowledge distillation technique to predict user device locations using a complex teacher model trained on a large-scale MIMO dataset, which was used to guide a more streamlined student model.

D. Green codecs

Green coding technology reduces energy consumption and bandwidth usage by optimising the data coding and decoding process. Efficient codecs can significantly reduce the amount of data transferred and increase transmission efficiency, thereby reducing overall energy consumption. Liao et al [31] proposed a deep learning based sphere decoding (SD) scheme to reduce the detection complexity of MIMO communication systems. Nguyue et al [32] proposed the fast deep learning-aided SD (FDL-SD) scheme, which achieves significant complexity reduction while fully preserving the performance of traditional SD. Liao et al [33] proposed a breadth-first SD scheme based on deep learning to reduce the detection complexity of MIMO communication systems.

III. Challenges and Future Work

A. Intelligent scheduling

1) Dynamic resource allocation: While AI-driven dynamic resource allocation offers significant potential for energy savings, challenges remain in developing algorithms that are

robust enough to handle the unpredictable and highly dynamic nature of 6G networks. Further research is needed to address the trade-off between optimal resource utilization and the computational overhead of the AI algorithms themselves. The scalability of these algorithms to massive networks with millions of devices also needs to be investigated. Developing more efficient and adaptable AI models that can learn and respond to network changes in real-time is crucial.

2) Mobile Edge Computing and Offloading: The effectiveness of MEC and offloading depends heavily on the accurate prediction of task characteristics and network conditions. Developing more precise prediction models is crucial for efficient offloading decisions. Furthermore, security and privacy concerns associated with offloading sensitive data to edge servers need to be addressed through robust security mechanisms. Research into energy-efficient edge server architectures and optimized offloading algorithms is also vital for maximizing the benefits of MEC.

B. Distributed Optimization

The main challenge for distributed optimization, particularly federated learning in 6G, lies in balancing the benefits of distributed training (privacy, reduced data transfer) with the communication overhead and computational complexities involved. Heterogeneity in the participating devices' capabilities poses a significant hurdle. Ensuring convergence speed and model accuracy while managing device heterogeneity and communication constraints needs further investigation. Developing efficient aggregation and communication protocols specifically optimized for 6G's unique characteristics is an important area of future research. Also, robust mechanisms to handle device failures and network disruptions within the distributed learning framework are needed.

C. Lightweight technology

1) Data Compression: While lossless compression techniques are well-established, further research into highly efficient lossy compression algorithms tailored for 6G applications (where some data loss may be acceptable) is needed. Exploring novel compression methods that go beyond traditional bit-level compression, such as semantic compression, holds significant promise but requires significant advancements in understanding and representing meaning effectively. The design of efficient compression and decompression algorithms that work in resource-constrained environments remains a challenge.

2) Lightweight for Model: Finding the optimal balance between model size reduction and accuracy loss is a crucial research direction. Developing more sophisticated model compression techniques that effectively reduce model size without sacrificing performance needs exploration. Furthermore, efficient methods for deploying and managing compressed models in dynamic 6G environments require further investigation. Research into model compression techniques that are compatible with various AI model architectures is necessary.

D. Green codecs

Developing codecs that are not only energy-efficient but also maintain high data rates and low latency in 6G networks is paramount. Research should focus on creating codecs that adapt dynamically to changing network conditions and user demands. Exploration of novel coding schemes inspired by advancements in deep learning and other AI methods could significantly improve the energy efficiency of data transmission. The complexity of these green codecs and their implementation in hardware should be carefully considered.

IV. Conclusion

This review has explored the crucial role of green communication technologies in enabling a sustainable 6G future. The escalating energy consumption associated with the high data rates, low latency, and massive connectivity promised by 6G necessitates a paradigm shift towards energy-efficient network design and operation. We have examined key enabling technologies, including intelligent scheduling (with a focus on dynamic resource allocation and Mobile Edge Computing), distributed optimization through Federated Learning, lightweight techniques such as data and model compression, and energy-efficient green codecs. Each technology category offers significant potential for reducing energy consumption, but also presents unique challenges. Trade-offs between performance and energy efficiency, the complexities of distributed systems, and the need for adaptable solutions in heterogeneous 6G environments remain significant hurdles. Future research should prioritize the development of more sophisticated algorithms that can dynamically adapt to changing network conditions while minimizing energy consumption without compromising performance.

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